# SME206
# Signals and Systems

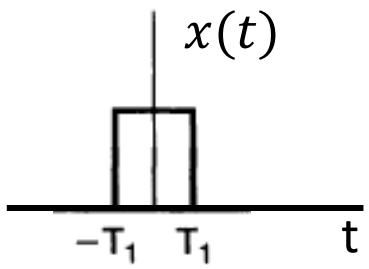
# 5. The Discrete-Time Fourier Transform


## LI Jiamin

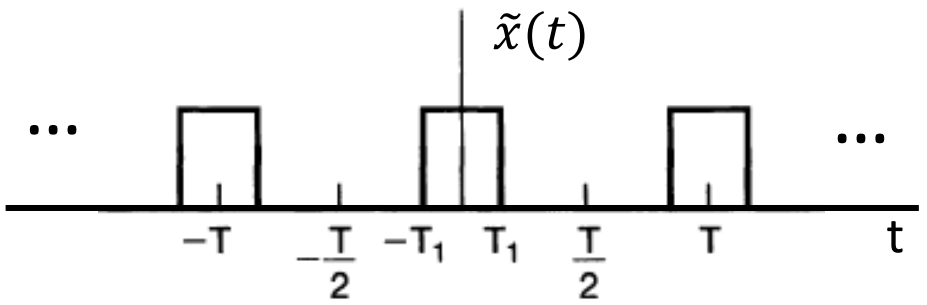
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# Previously in Chapter 4…

- How we derived the CT Fourier Transform?




periodically padded

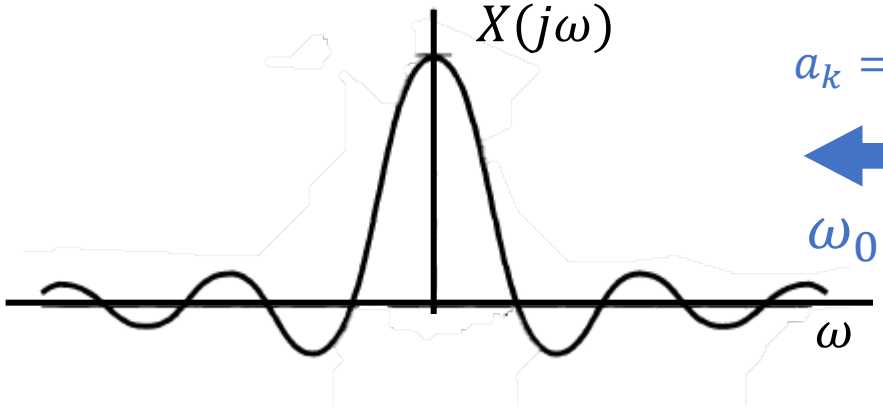


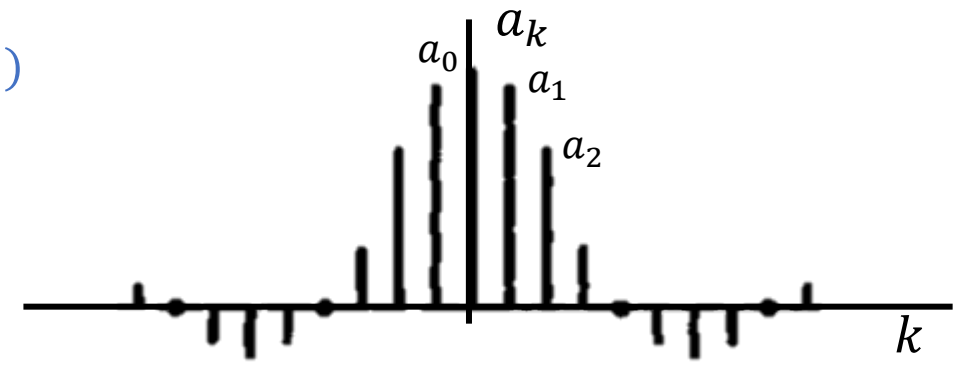

Fourier Series Analysis

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-jk\omega_0 t}dt$$

$$a_k = \frac{1}{T}X(jk\omega_0)$$

$\omega_0 \to 0$

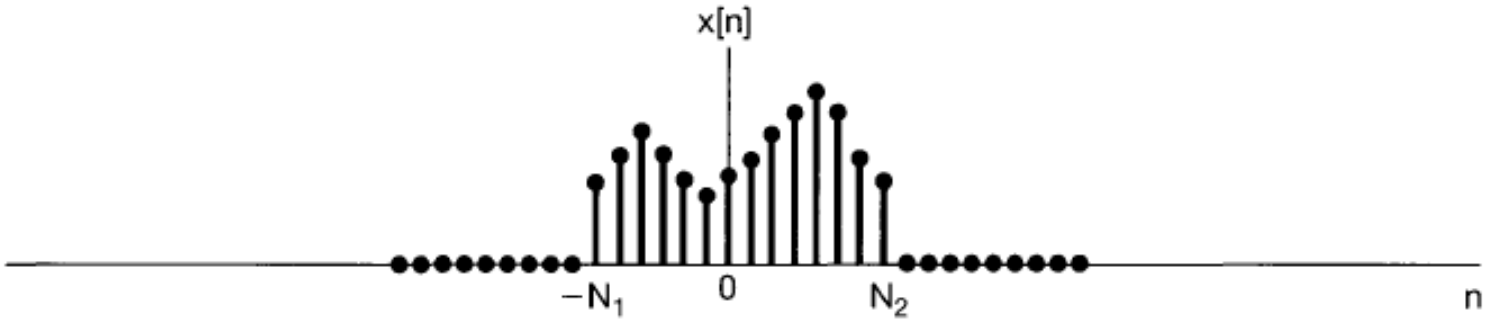


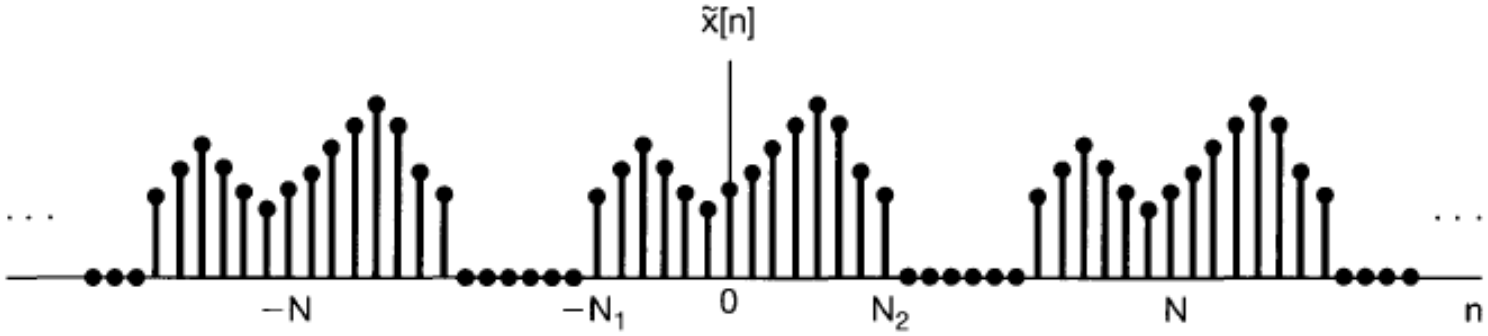

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}, where\ a_k = \frac{1}{T}\int_T \tilde{x}(t)e^{-jk\omega_0 t}dt$$

# DT Fourier Transform: Aperiodic Signals

- **Derivation** for an aperiodic DT signal $x[n]$:




1) Construct a periodic signal $\tilde{x}[n]$ from the finite-duration signal $x[n]$



# DT Fourier Transform: Aperiodic Signals

2) For the periodic signal $\tilde{x}[n]$, apply DT Fourier series:

$$\tilde{x}[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}, where\ a_k = \frac{1}{N} \sum_{k=<N>} \tilde{x}[n] e^{-jk\omega_0 n}$$

- Consider summing over a period that includes $-N_1 \le n \le N_2$

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} \tilde{x}[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}$$

- *Define:* $\boldsymbol{X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}}$

- *Then:* $\boldsymbol{a_k = \frac{1}{N} X(e^{jk\omega_0})}, \quad where\ \omega_0 = \frac{2\pi}{N}$

# DT Fourier Transform: Aperiodic Signals

3) Substitute $a_k = \frac{1}{N} X\left(e^{jk\omega_0}\right)$ into the Fourier series representation of $\tilde{x}[n]$

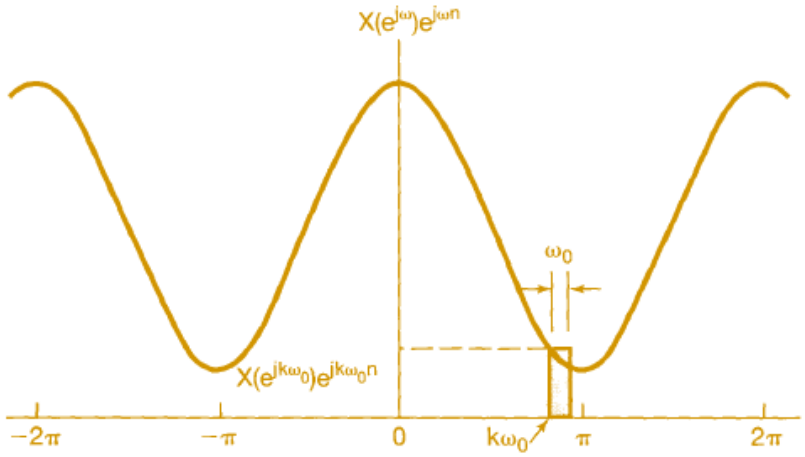
$$\tilde{x}[n] = \sum_{k=<N>} \left[\frac{1}{N} X\left(e^{jk\omega_0}\right)\right] e^{jk\omega_0 n} = \frac{1}{2\pi} \sum_{k=<N>} X\left(e^{jk\omega_0}\right) e^{jk\omega_0 n} \omega_0 \qquad (\omega_0 = \frac{2\pi}{N})$$

- As $N \to \infty$, $\omega_0 \to 0$, and $\tilde{x}[n] \to x[n]$

$$\tilde{x}[n] = \frac{1}{2\pi} \sum_{k=<N>} X\left(e^{jk\omega_0}\right) e^{jk\omega_0 n} \omega_0 \;\to\; \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

- Therefore:

$$\boldsymbol{x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega}$$



# DT Fourier Transform: Aperiodic Signals

- Why is $X(e^{j\omega})e^{j\omega n}$ ***periodic***?
- $e^{j\omega n}$: periodic with period $2\pi$
- $X(e^{j\omega})$:

$$X\left(e^{j(\omega+2\pi)}\right) = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega+2\pi)n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}(e^{-j2\pi})^n$$

  - As $(e^{-j2\pi})^n = 1^n = 1$ for any integer $n$, $X(e^{j\omega})$ is periodic with period $2\pi$.
- Therefore, the product $\boldsymbol{X(e^{j\omega})e^{j\omega n}}$ **is periodic with period $2\pi$**.

Q. How about the **CT Fourier Transform**?

- $X\left(j(\omega+2\pi)\right) = \int_{-\infty}^{+\infty} x(t)e^{-j(\omega+2\pi)t}dt = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}(e^{-j2\pi})^t dt, \quad t \in R$
- Thus, $X\left(j(\omega+2\pi)\right) \neq X(j\omega)$

# DT Fourier Transform: Aperiodic Signals

- **Discrete-Time Fourier Transform Pair:**

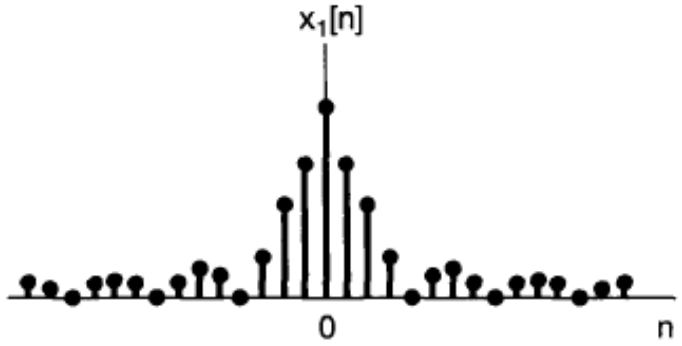
$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega \quad \text{(synthesis equation)}$$

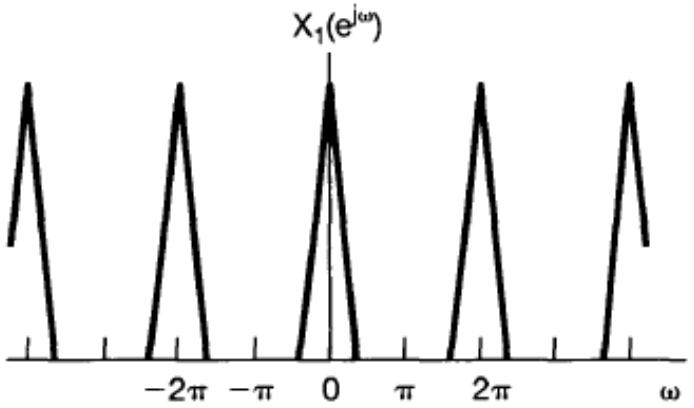
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \text{(analysis equation)}$$

- As $X(e^{j\omega})e^{j\omega n}$ is periodic with period $2\pi$, *the interval of integration* can be taken as ***any*** *interval of length* $2\pi$.
- The Fourier transform $X(e^{j\omega})$ is often referred to as the *spectrum of* $x[n]$.

# DT Fourier Transform: Aperiodic Signals

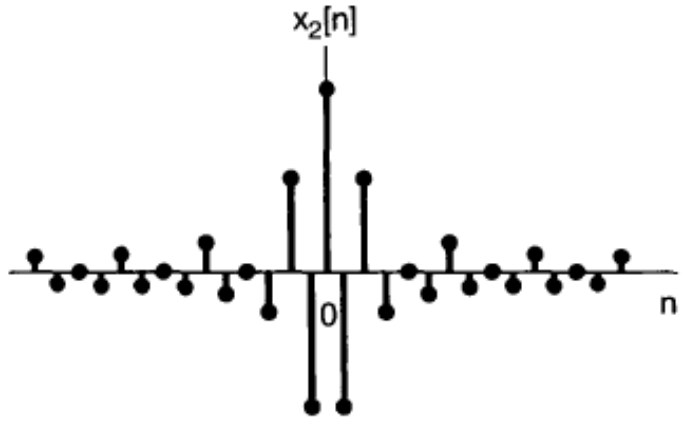
- **Discrete-Time Fourier Transform Pair**
  - Periodicity of $2\pi$.

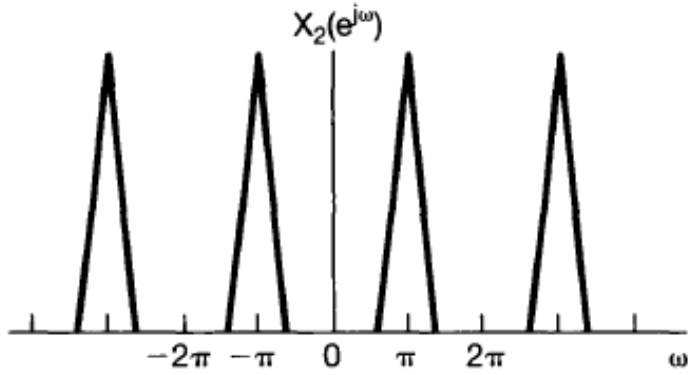



Frequencies near 0 or even multiples of $\pi$:

***Lower frequency signals***




Frequencies near odd multiples of $\pi$:

***Higher frequency signals***

# DT Fourier Transform: Aperiodic Signals

Q. Consider the DT signal $x[n] = a^n u[n],\ |a| < 1$

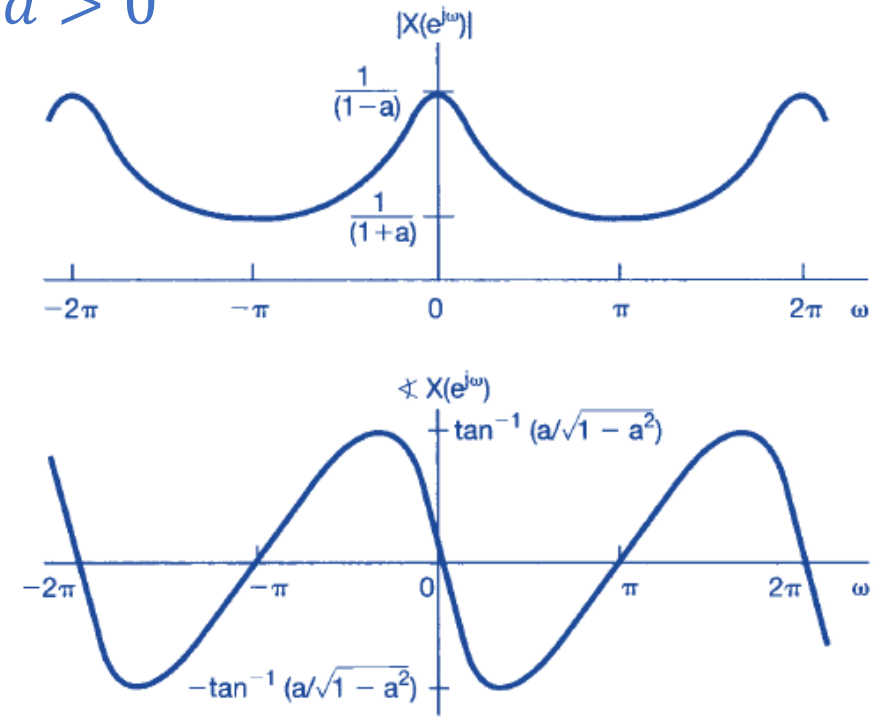
Determine the Fourier transform.

(Example 5.1)

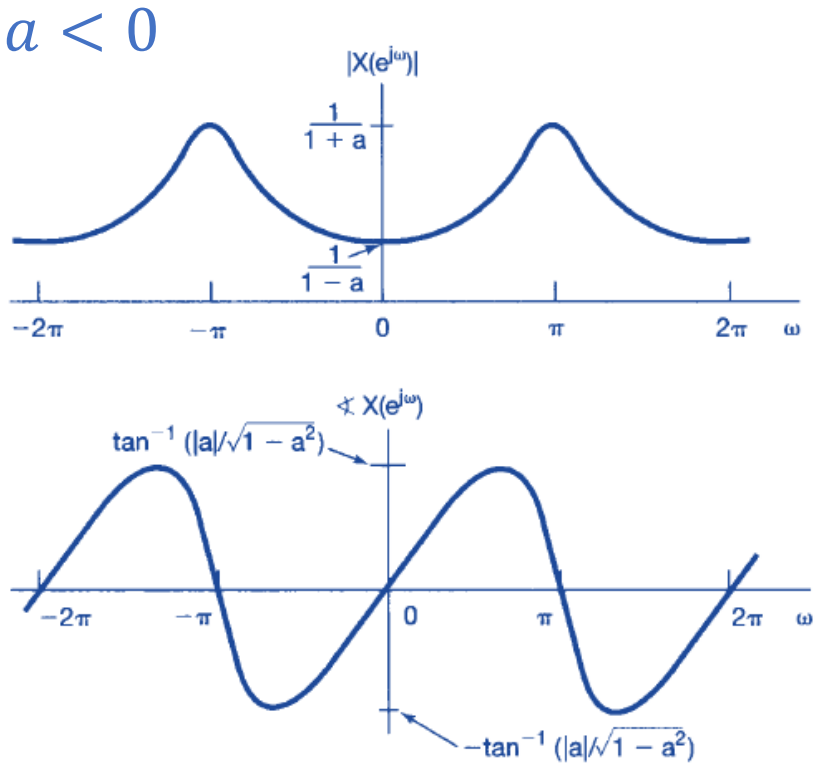
$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

Self-Practice:
(Example 5.2)

$a > 0$




$a < 0$



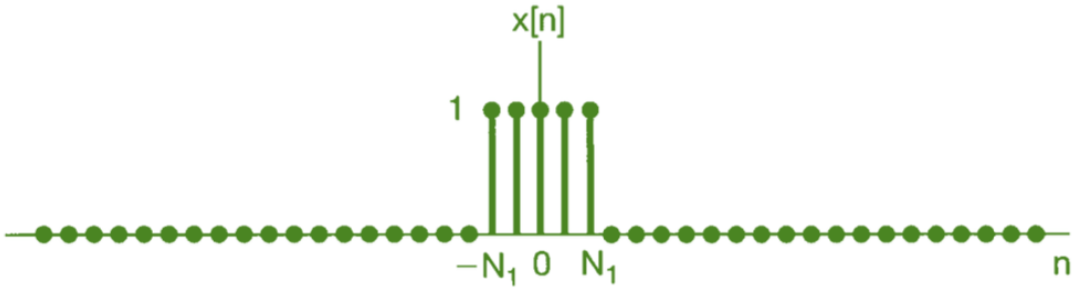
# DT Fourier Transform: Aperiodic Signals

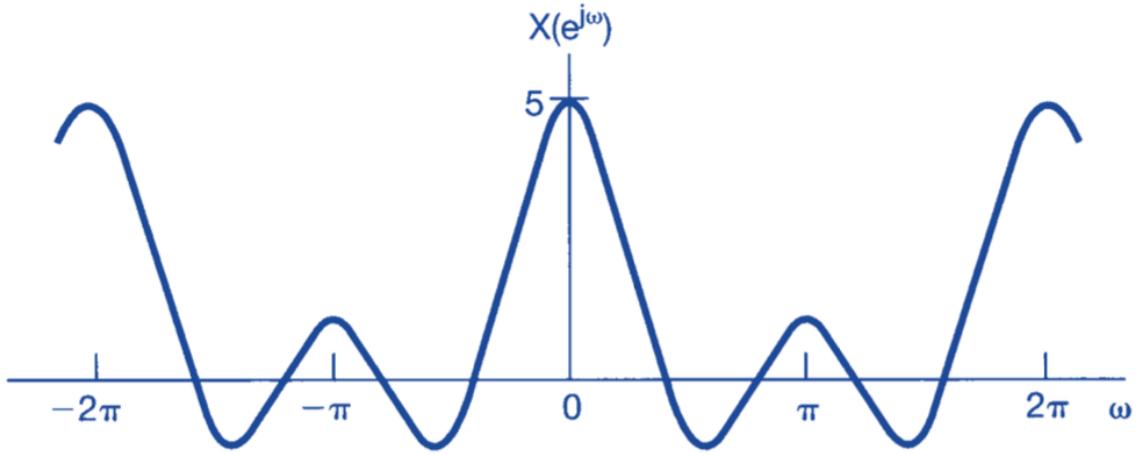
Q. Consider the rectangular pulse

(Example 5.3)

$$x[n] = \begin{cases} 1, & |n| \le N_1 \\ 0, & |n| > N_1 \end{cases}$$

which is illustrated below for $N_1 = 2$. Find the DT Fourier transform $X(e^{j\omega})$.




Periodic! ⟵ $X(e^{j\omega}) = \dfrac{sin\omega(N_1 + 1/2)}{\sin(\omega/2)}$

(unlike sinc function in the CT case)

# DT Fourier Transform: Periodic Signals

- **DT periodic signals: Fourier transform vs. Fourier series**
- Consider the signal $\boldsymbol{x[n] = e^{j\omega_0 n}}$,
- The Fourier transform of $x[n]$ is the *impulse train*:

$$\boldsymbol{X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)}$$

- **Verification?**
- By evaluating its inverse FT:

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega = \frac{1}{2\pi}\int_{2\pi}\sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)\, e^{j\omega n}d\omega$$

- Note that any interval of length $2\pi$ includes exactly one impulse in the summation → $x[n] = e^{j\omega_0 n}$

# DT Fourier Transform: Periodic Signals

- **DT periodic signals: Fourier transform vs. Fourier series**
- Consider a periodic sequence $x[n]$ with period $N$, and with FS representation:

$$x[n] = \sum_{k=<N>} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

- Fourier transform:

$$\boldsymbol{X\left(e^{j\omega}\right) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi k}{N}\right)}$$

- Verification?

# DT Fourier Transform: Periodic Signals

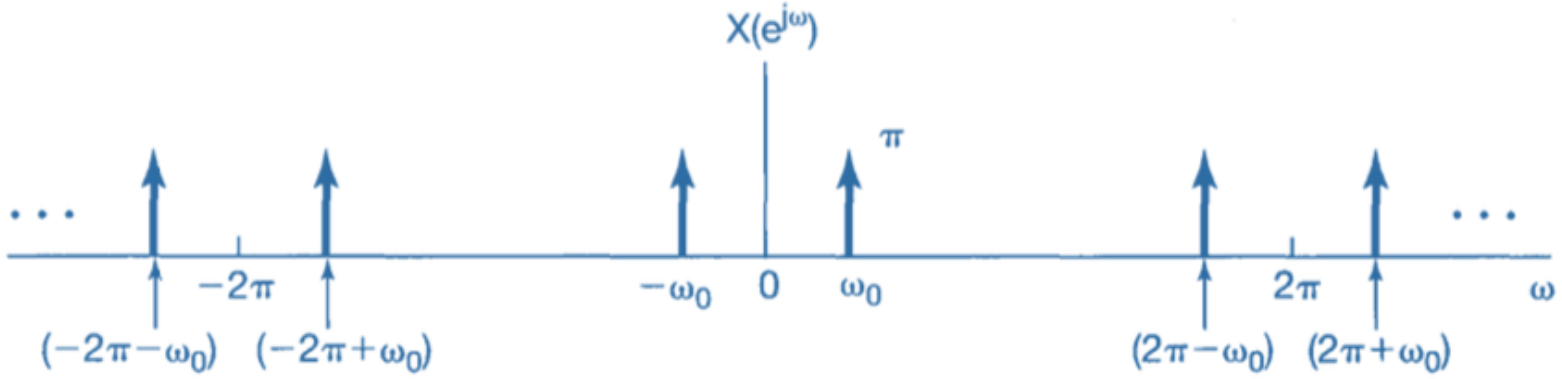
(Example 5.5)

Q. Consider the periodic signal

$$x[n] = cos\omega_0 n = \frac{1}{2}e^{j\omega_0 n} + \frac{1}{2}e^{-j\omega_0 n}, \qquad \text{with } \omega_0 = \frac{2\pi}{5}$$

Find the DT Fourier transform.

- $X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \pi\delta(\omega - \frac{2\pi}{5} - 2\pi l) + \sum_{l=-\infty}^{\infty} \pi\delta(\omega + \frac{2\pi}{5} - 2\pi l)$
- Or, $X(e^{j\omega}) = \pi\delta\left(\omega - \frac{2\pi}{5}\right) + \pi\delta\left(\omega + \frac{2\pi}{5}\right), \ -\pi \le \omega < \pi,$ where $X(e^{j\omega})$ repeats periodically with a period of $2\pi$.



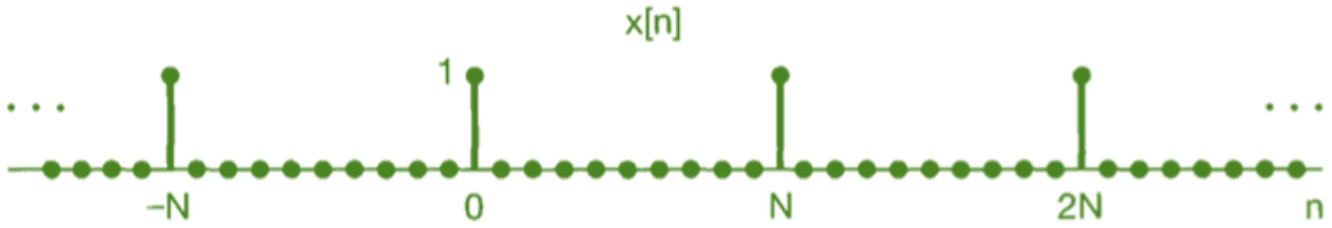
# DT Fourier Transform: Periodic Signals

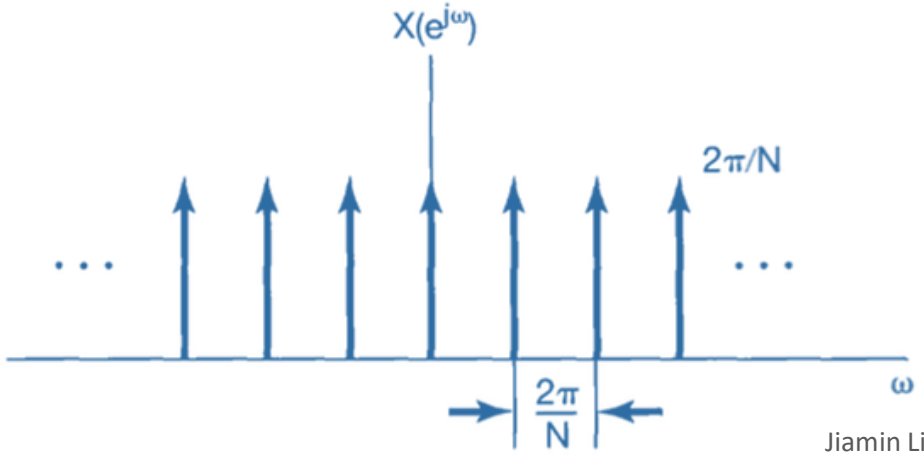
(Example 5.6)

Q. Consider a periodic impulse train

$$x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

Find the DT Fourier transform.




$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

# DT Fourier Transform: Convergence

- The *analysis equation* will converge
  - if $x[n]$ is absolutely absolutely summable: $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$
  - Or if $x[n]$ has finite energy: $\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$

- There are *generally no convergence issue* associated with the *synthesis equation*, since the integral is over a finite interval of integration.

# Properties of DT Fourier Transform

- Recall for DT signals:
  - Fourier Transform: $$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$
  - Inverse Fourier Transform: $$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

- **Notation:**

$$\boldsymbol{x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})}$$

$$\boldsymbol{\mathcal{F}\{x[n]\} = X(e^{j\omega})}$$

$$\boldsymbol{\mathcal{F}^{-1}\{X(e^{j\omega})\} = x[n]}$$

# Properties of DT Fourier Transform

- **Periodicity**
- The discrete-time Fourier transform is *always periodic in $\omega$ with period $2\pi$*

$$X\left(e^{j(\omega+2\pi)}\right) = X\left(e^{j\omega}\right)$$

- This is *in contrast to the CT FT*, which in general is *not periodic*.

# Properties of DT Fourier Transform

- **Linearity**
- If

$$x_1[n] \overset{\mathcal{F}}{\longleftrightarrow} X_1(e^{j\omega})$$
$$x_2[n] \overset{\mathcal{F}}{\longleftrightarrow} X_2(e^{j\omega})$$

- Then

$$\boldsymbol{ax_1[n] + bx_2[n] \overset{\mathcal{F}}{\longleftrightarrow} aX_1(e^{j\omega}) + bX_2(e^{j\omega})}$$

- Proof?

# Properties of DT Fourier Transform

- **Time Shifting & Frequency Shifting**
- If

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$$

- Then

$$\boldsymbol{x[n-n_0] \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega n_0} X(e^{j\omega})}$$

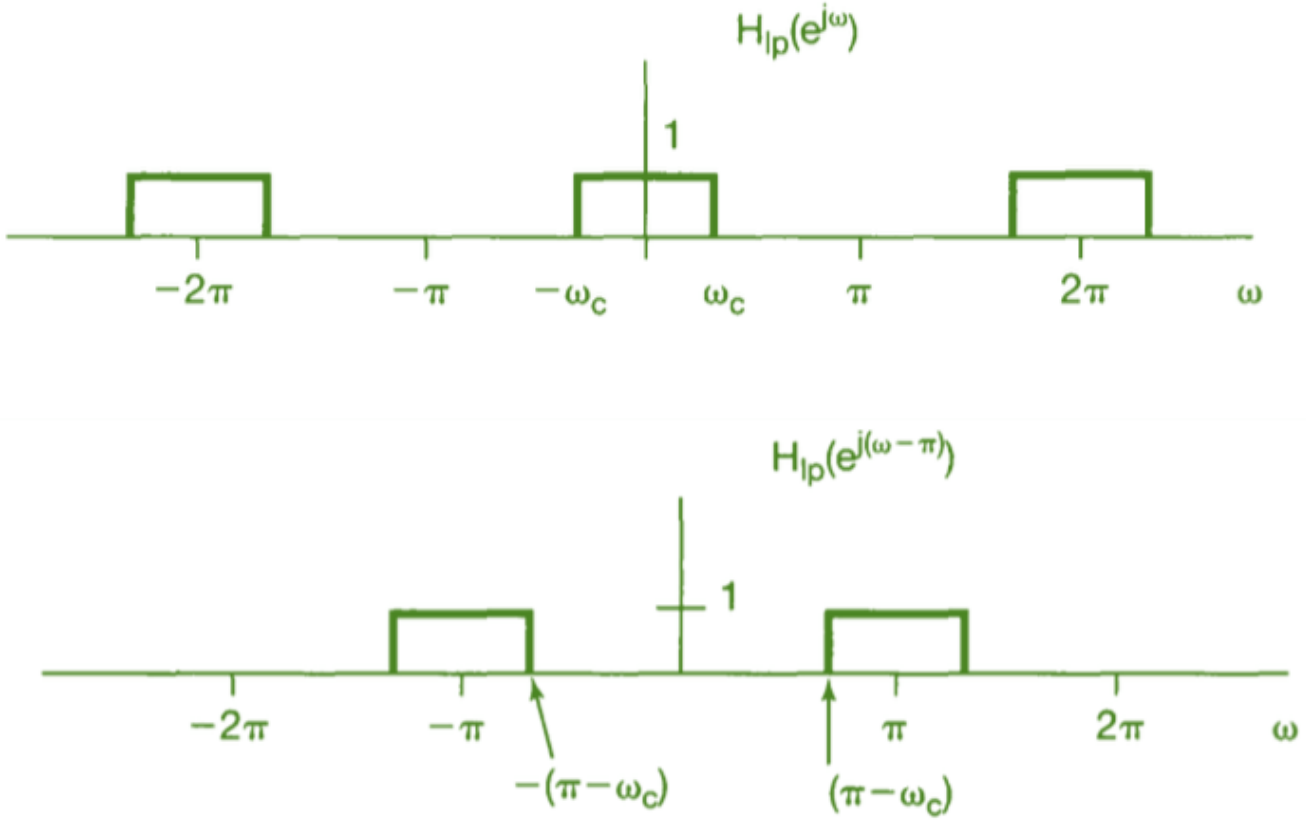
$$\boldsymbol{e^{j\omega_0 n} x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j(\omega-\omega_0)})}$$

- Proof?

# Properties of DT Fourier Transform

- Observation: a special relationship between ideal lowpass and ideal highpass *discrete-time filters*.

(Example 5.7)




$$H_{hp}(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})$$

$$\downarrow$$

$$h_{hp}[n] = e^{j\pi n} h_{lp}[n]$$

$$= (-1)^n h_{lp}[n]$$

# Properties of DT Fourier Transform

- **Conjugation and Conjugate Symmetry**
- If

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$$

- Then

$$x^*[n] \overset{\mathcal{F}}{\longleftrightarrow} X^*(e^{-j\omega})$$

Proof?

- If $x[n]$ is real (*i.e.*, $x[n] = x^*[n]$), then $X(e^{j\omega}) = X^*(e^{-j\omega})$
  - $Re\{X(e^{j\omega})\}$: an even function of $\omega$
  - $Im\{X(e^{j\omega})\}$: an odd function of $\omega$
  - $|X(e^{j\omega})|$: an even function of $\omega$
  - $\measuredangle X(e^{j\omega})$: an odd function of $\omega$

# Properties of DT Fourier Transform

- **Conjugation and Conjugate Symmetry**
- If

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$$

- Then

$$x^*[n] \overset{\mathcal{F}}{\longleftrightarrow} X^*(e^{-j\omega})$$

Proof?

- If $x[n]$ is real (*i.e.*, $x[n] = x^*[n]$), then $X(e^{j\omega}) = X^*(e^{-j\omega})$
  - Express $x[n]$ in terms of the sum of an even function and an odd function
    - $Ev\{x[n]\} \overset{\mathcal{F}}{\longleftrightarrow} Re\{X(e^{j\omega})\}$
    - $Od\{x[n]\} \overset{\mathcal{F}}{\longleftrightarrow} jIm\{X(e^{j\omega})\}$

# Properties of DT Fourier Transform

- **Differencing & Accumulation**
- If

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$$

- Then

$$\boldsymbol{x[n]-x[n-1] \overset{\mathcal{F}}{\longleftrightarrow} (1-e^{-j\omega})X(e^{j\omega})}$$

$$\boldsymbol{\sum_{m=-\infty}^{n} x[m] \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)}$$

# Properties of DT Fourier Transform

Q. Given that: $g[n] = \delta[n] \overset{\mathcal{F}}{\longleftrightarrow} G(e^{j\omega}) = 1$,

find the FT $X(e^{j\omega})$ of the unit step $x[n] = u[n]$.

(Example 5.8)

$$X(e^{j\omega}) = \frac{1}{1 - e^{-j\omega}} G(e^{j\omega}) + \pi G(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$$

$$= \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$$

# Properties of DT Fourier Transform

- **Time Reversal**
- If

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$$

- Then

$$\boldsymbol{x[-n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{-j\omega})}$$

- For real-valued even signals (where $x[n] = x[-n]$):

$$X(e^{j\omega}) = \mathcal{F}\{x[n]\} = \mathcal{F}\{x[-n]\} = X(e^{-j\omega})$$

  - Together with the conjugation property,

→ DT FT of *real & even* signals is also *real & even*.

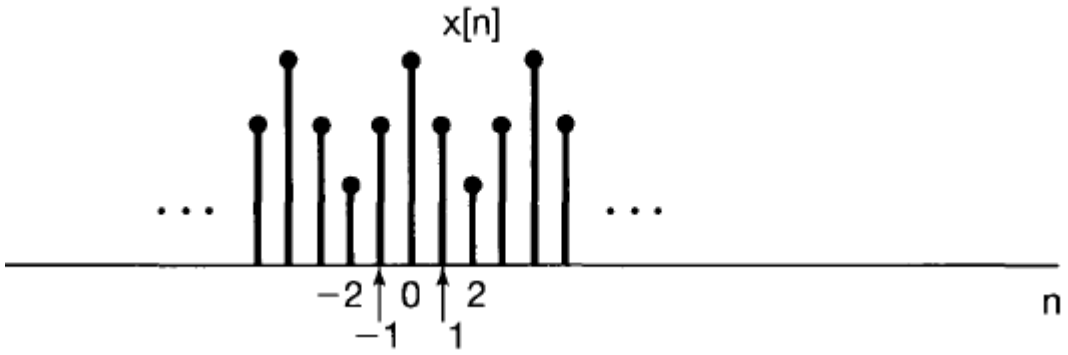
# Properties of DT Fourier Transform

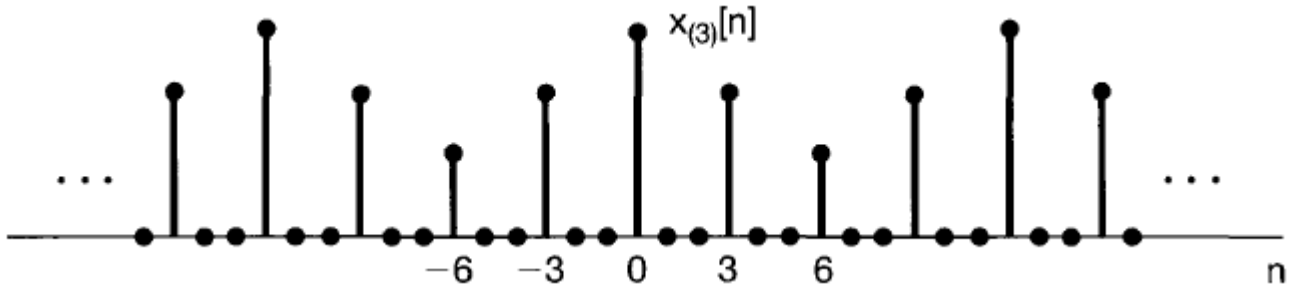
- **Time Expansion**
- Let $k$ be a positive integer, and define the signal

$$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{if } n \text{ is not a multiple of } k \end{cases}$$

- Then

$$x_{(k)}[n] \xleftrightarrow{\mathcal{F}} X(e^{jk\omega})$$




(obtained from $x[n]$ by placing $k-1$ zeros between successive values of the original signal)

# Properties of DT Fourier Transform

- **Time Expansion**
- Let $k$ be a positive integer, and define the signal

$$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{if } n \text{ is not a multiple of } k \end{cases}$$

- Then

$$\boldsymbol{x_{(k)}[n] \overset{\mathcal{F}}{\longleftrightarrow} X\left(e^{jk\omega}\right)}$$

- Proof ($x_{(k)}$[n] equals 0 unless $n = rk, r \in Z$)

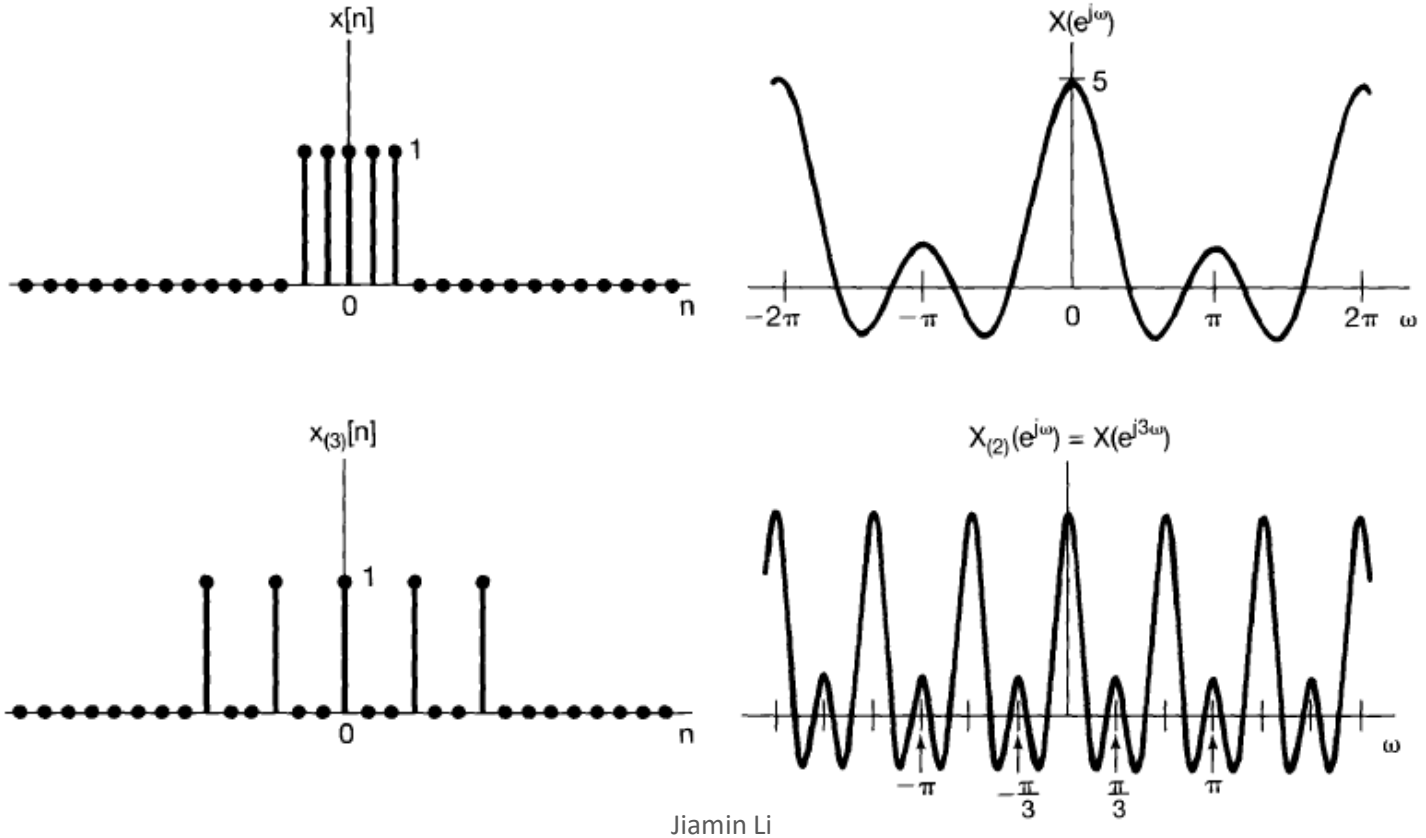
$$X_{(k)}\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} x_{(k)}[n]e^{-j\omega n} = \sum_{r=-\infty}^{\infty} x_{(k)}[rk]e^{-j\omega rk} = \sum_{r=-\infty}^{\infty} x[r]e^{-j(k\omega)r} = X\left(e^{jk\omega}\right)$$

# Properties of DT Fourier Transform

- **Time Expansion**

$$x_{(k)}[n] \xleftrightarrow{\mathcal{F}} X\left(e^{jk\omega}\right)$$

- Fourier transform *compressed:* $X\left(e^{jk\omega}\right)$ is periodic with period $2\pi/k$

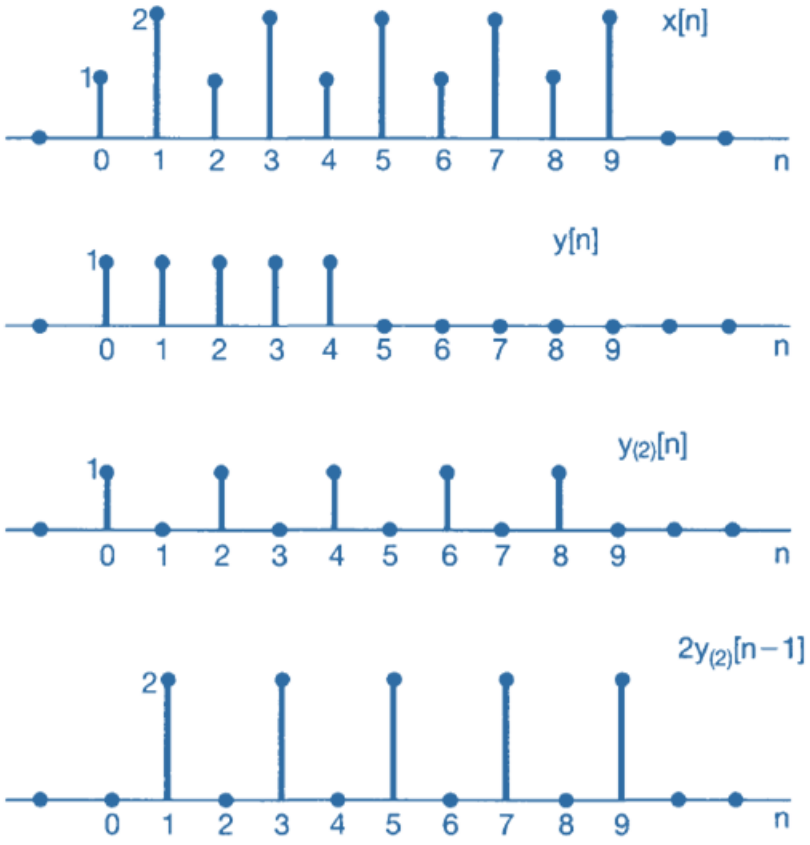


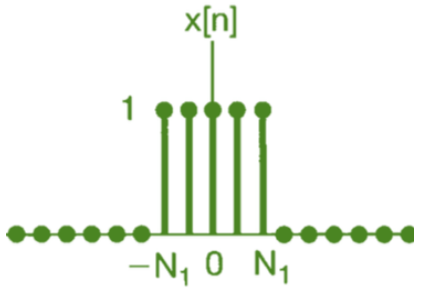
# Properties of DT Fourier Transform

Q. Consider the sequence $x[n]$, which can be related to the simpler sequence $y[n]$:

$$x[n] = y_{(2)}[n] + 2y_{(2)}[n-1]$$ (Example 5.9)

Find the DT Fourier transform of $x[n]$.




*Recall:*

$$X(e^{j\omega}) = \frac{sin\omega(N_1 + 1/2)}{\sin(\omega/2)}$$

$$Y(e^{j\omega}) = e^{-j2\omega}\frac{\sin(5\omega/2)}{\sin(\omega/2)} \qquad y_{(2)}[n] \overset{\mathcal{F}}{\longleftrightarrow} e^{-j4\omega}\frac{\sin(5\omega)}{\sin(\omega)}$$

$$2y_{(2)}[n-1] \overset{\mathcal{F}}{\longleftrightarrow} 2e^{-j5\omega}\frac{\sin(5\omega)}{\sin(\omega)}$$

$$X(e^{j\omega}) = e^{-j4\omega}(1 + 2e^{-j\omega})\left(\frac{\sin(5\omega)}{\sin(\omega)}\right)$$

# Properties of DT Fourier Transform

- **Differentiation in Frequency**
- If $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$
- Then

$$\boldsymbol{nx[n] \overset{\mathcal{F}}{\longleftrightarrow} j\frac{dX(e^{j\omega})}{d\omega}}$$

- Proof

$$\frac{d}{d\omega}X(e^{j\omega}) = \frac{d}{d\omega}\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty}(-jn)x[n]e^{-j\omega n}$$

$$\Rightarrow \mathcal{F}\{nx[n]\} = j\frac{dX(e^{j\omega})}{d\omega}$$

# Properties of DT Fourier Transform

- **Parseval's Relation**
- If $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega})$
- Then

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

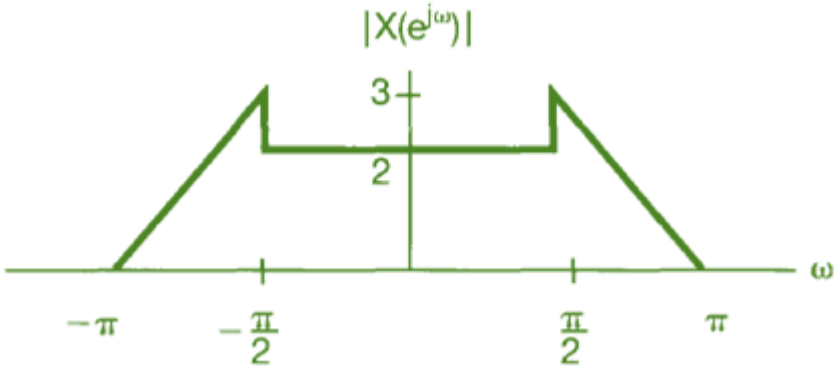
- **LHS**: total energy in the signal $x[n]$
- **RHS**: Integrating energy per unit frequency ($|X(e^{j\omega})|^2/2\pi$) over a full $2\pi$ interval *of distinct DT frequencies*
- $|X(e^{j\omega})|^2$: energy-density spectrum of the signal $x[n]$

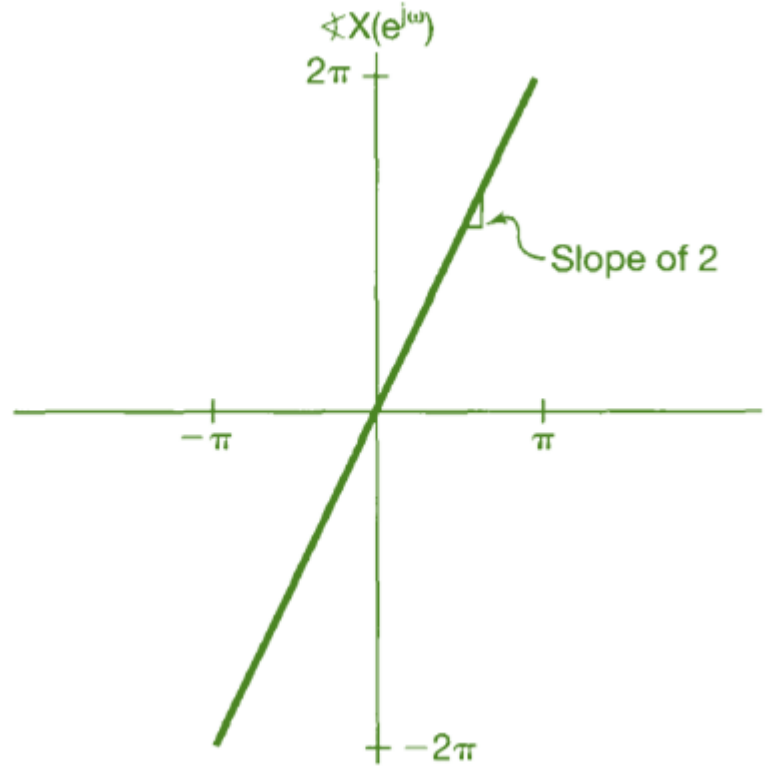
# Properties of DT Fourier Transform

Q. Consider the sequence $x[n]$ whose FT $X(e^{j\omega})$ is depicted for $-\pi \leq \omega \leq \pi$.

Determine whether or not, in time domain, $x[n]$ is

1) periodic
2) real
3) even
4) of finite energy



# Properties of DT Fourier Transform

- **Convolution**

$$y[n] = x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

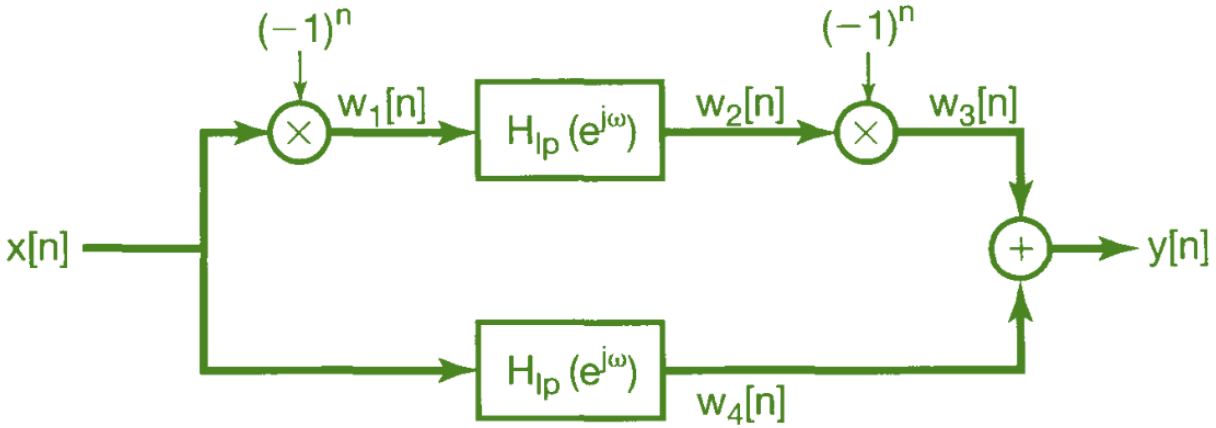
where $X(e^{j\omega})$, $H(e^{j\omega})$, and $Y(e^{j\omega})$ are the FTs of $x[n]$, $h[n]$, and $y[n]$.

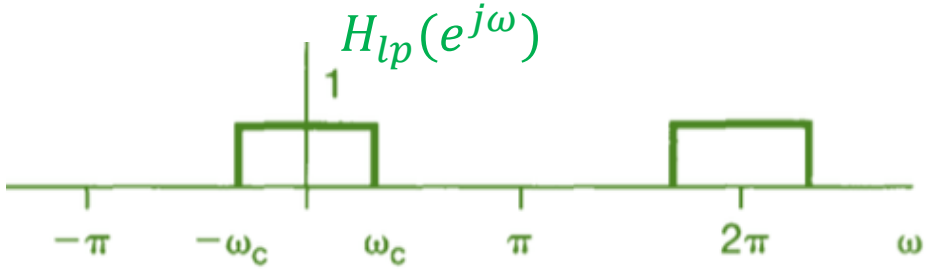
- *Frequency response* of a DT LTI system **is** the *FT of its impulse response.*

# Properties of DT Fourier Transform

(Example 5.14)

Q. Consider the system with input $x[n]$ and output $y[n]$. The LTI systems with frequency response $H_{lp}(e^{j\omega})$ are ideal lowpass filters with cutoff frequency $\pi/4$ and unity gain in the passband. Find the overall system frequency response.

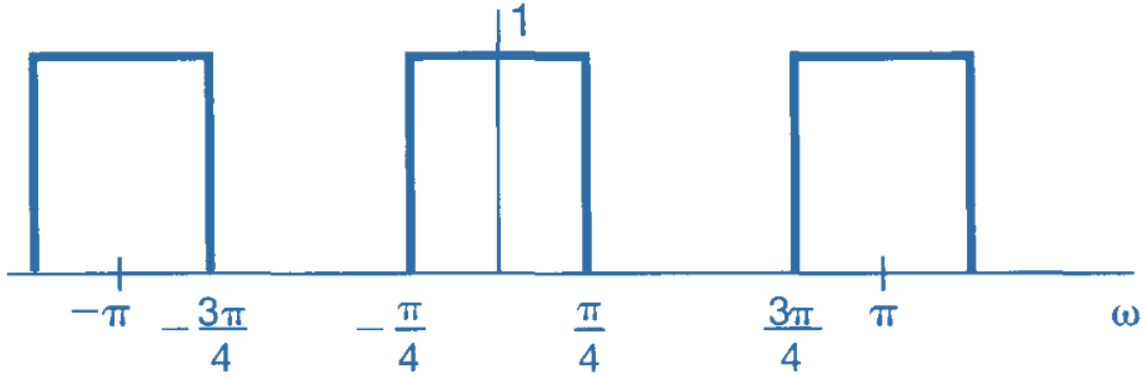



Ideal low-pass filter

- $W_1[e^{j\omega}]$
- $W_2[e^{j\omega}]$
- $W_3[e^{j\omega}]$
- $W_4[e^{j\omega}]$
- $Y[e^{j\omega}]$

$H(e^{j\omega})$ Ideal band-stop filter



# Properties of DT Fourier Transform

- **Multiplication**

*periodic convolution*

$$y[n] = x_1[n]x_2[n] \overset{\mathcal{F}}{\longleftrightarrow} Y(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)})d\theta$$

where $X_1(e^{j\omega})$, $X_2(e^{j\omega})$ and $Y(e^{j\omega})$ are the FTs of $x_1[n]$, $x_2[n]$ and $y[n]$.

- Proof?

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_1[n]x_2[n]e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x_2[n]\left\{\frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})e^{j\theta n}d\theta\right\}e^{-j\omega n} = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})\left[\sum_{n=-\infty}^{\infty} x_2[n]\,e^{-j(\omega-\theta)n}\right]d\theta$$

$$= \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)})\,d\theta$$

# Properties of DT Fourier Transform

(Example 5.15)

Q. Consider $x[n] = x_1[n]x_2[n]$, where $x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}$, $x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$.

Find the Fourier transform of $x[n]$.

$$X(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta$$

- Convert periodic convolution to the ordinary one (aperiodic convolution): define

$$\widehat{X_1}(e^{j\omega}) = \begin{cases} X_1(e^{j\omega}), & for\ -\pi < \omega < \pi \\ 0, & otherwise \end{cases}$$

- Thus,

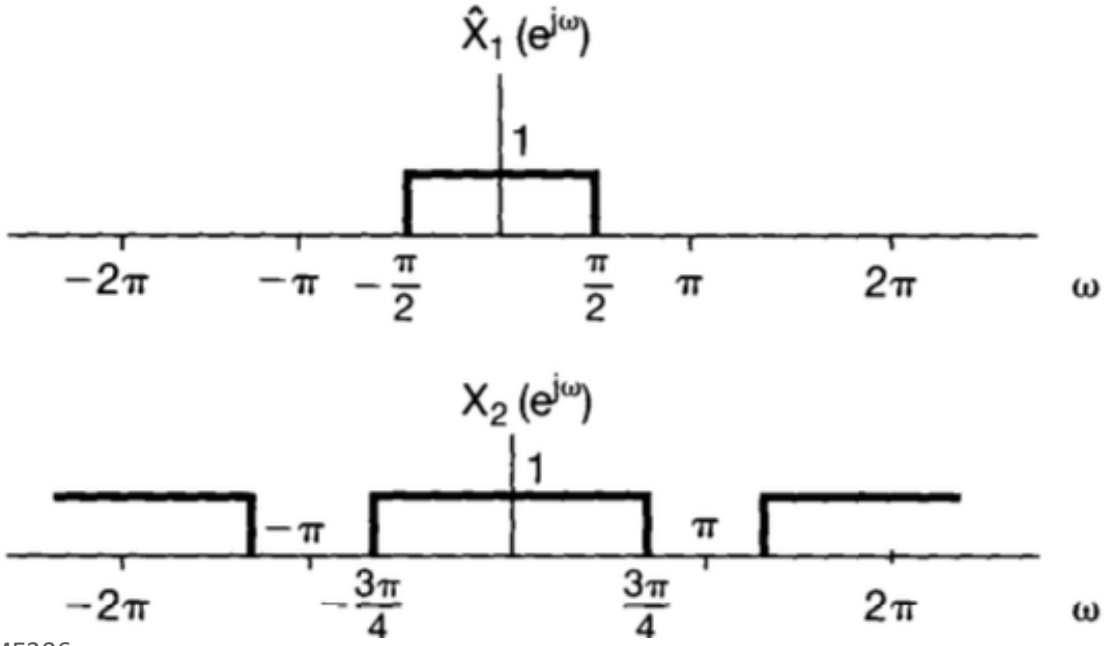
$$X(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{X_1}(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{X_1}(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta$$

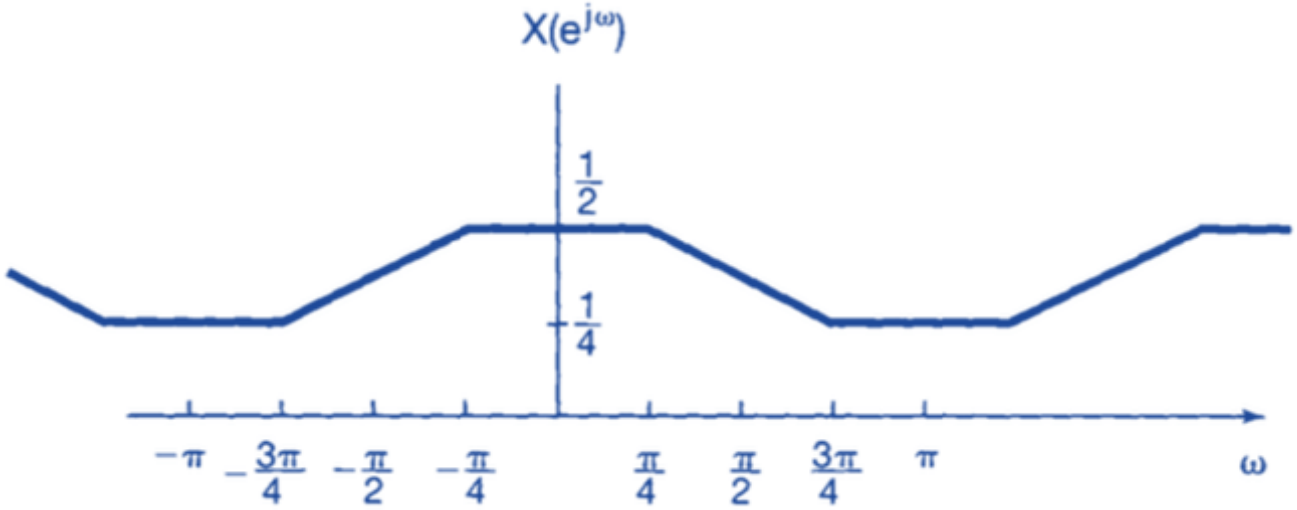
# Properties of DT Fourier Transform

Q. Consider $x[n] = x_1[n]x_2[n]$, where $x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}$, $x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$.

Find the Fourier transform of $x[n]$.

$$X(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{X_1}(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{X_1}(e^{j\omega})X_2(e^{j(\omega-\theta)})d\theta$$



# Properties of DT Fourier Transform

- Table of Fourier Transform Properties: Table 5.1
- Table of Basic Fourier Transform Pairs: Table 5.2
- Duality

**TABLE 5.1** PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

| Section | Property | Aperiodic Signal | Fourier Transform |
|---|---|---|---|
| | | $x[n]$<br>$y[n]$ | $X(e^{j\omega})$<br>$Y(e^{j\omega})$ periodic with period $2\pi$ |
| 5.3.2 | Linearity | $ax[n] + by[n]$ | $aX(e^{j\omega}) + bY(e^{j\omega})$ |
| 5.3.3 | Time Shifting | $x[n - n_0]$ | $e^{-j\omega n_0}X(e^{j\omega})$ |
| 5.3.3 | Frequency Shifting | $e^{j\omega_0 n}x[n]$ | $X(e^{j(\omega-\omega_0)})$ |
| 5.3.4 | Conjugation | $x^*[n]$ | $X^*(e^{-j\omega})$ |
| 5.3.6 | Time Reversal | $x[-n]$ | $X(e^{-j\omega})$ |
| 5.3.7 | Time Expansion | $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$ | $X(e^{jk\omega})$ |
| 5.4 | Convolution | $x[n] * y[n]$ | $X(e^{j\omega})Y(e^{j\omega})$ |
| 5.5 | Multiplication | $x[n]y[n]$ | $\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$ |
| 5.3.5 | Differencing in Time | $x[n] - x[n-1]$ | $(1 - e^{-j\omega})X(e^{j\omega})$ |
| 5.3.5 | Accumulation | $\sum_{k=-\infty}^{n} x[k]$ | $\frac{1}{1 - e^{-j\omega}}X(e^{j\omega})$<br>$+\pi X(e^{j0})\sum_{k=-\infty}^{+\infty}\delta(\omega - 2\pi k)$ |
| 5.3.8 | Differentiation in Frequency | $nx[n]$ | $j\frac{dX(e^{j\omega})}{d\omega}$ |
| 5.3.4 | Conjugate Symmetry for Real Signals | $x[n]$ real | $X(e^{j\omega}) = X^*(e^{-j\omega})$<br>$\mathcal{R}e\{X(e^{j\omega})\} = \mathcal{R}e\{X(e^{-j\omega})\}$<br>$\mathcal{I}m\{X(e^{j\omega})\} = -\mathcal{I}m\{X(e^{-j\omega})\}$<br>$\lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert$<br>$\measuredangle X(e^{j\omega}) = -\measuredangle X(e^{-j\omega})$ |
| 5.3.4 | Symmetry for Real, Even Signals | $x[n]$ real an even | $X(e^{j\omega})$ real and even |
| 5.3.4 | Symmetry for Real, Odd Signals | $x[n]$ real and odd | $X(e^{j\omega})$ purely imaginary and odd |
| 5.3.4 | Even-odd Decomposition of Real Signals | $x_e[n] = \mathcal{E}v\{x[n]\}$ $[x[n]$ real$]$<br>$x_o[n] = \mathcal{O}d\{x[n]\}$ $[x[n]$ real$]$ | $\mathcal{R}e\{X(e^{j\omega})\}$<br>$j\mathcal{I}m\{X(e^{j\omega})\}$ |
| 5.3.9 | Parseval's Relation for Aperiodic Signals | $\sum_{n=-\infty}^{+\infty}\lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi}\lvert X(e^{j\omega})\rvert^2 d\omega$ | |

**TABLE 5.2** BASIC DISCRETE-TIME FOURIER TRANSFORM PAIRS

| Signal | Fourier Transform | Fourier Series Coefficients (if periodic) |
|---|---|---|
| $\sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$ | $2\pi\sum_{k=-\infty}^{+\infty} a_k\delta\left(\omega - \frac{2\pi k}{N}\right)$ | $a_k$ |
| $e^{j\omega_0 n}$ | $2\pi\sum_{l=-\infty}^{+\infty}\delta(\omega - \omega_0 - 2\pi l)$ | (a) $\omega_0 = \frac{2\pi m}{N}$<br>$a_k = \begin{cases} 1, & k = m, m \pm N, m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$<br>(b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic |
| $\cos\omega_0 n$ | $\pi\sum_{l=-\infty}^{+\infty}\{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$ | (a) $\omega_0 = \frac{2\pi m}{N}$<br>$a_k = \begin{cases} \frac{1}{2}, & k = \pm m, \pm m \pm N, \pm m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$<br>(b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic |
| $\sin\omega_0 n$ | $\frac{\pi}{j}\sum_{l=-\infty}^{+\infty}\{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$ | (a) $\omega_0 = \frac{2\pi r}{N}$<br>$a_k = \begin{cases} \frac{1}{2j}, & k = r, r \pm N, r \pm 2N, \ldots \\ -\frac{1}{2j}, & k = -r, -r \pm N, -r \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$<br>(b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic |
| $x[n] = 1$ | $2\pi\sum_{l=-\infty}^{+\infty}\delta(\omega - 2\pi l)$ | $a_k = \begin{cases} 1, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ |
| Periodic square wave<br>$x[n] = \begin{cases} 1, & \lvert n\rvert \leq N_1 \\ 0, & N_1 < \lvert n\rvert \leq N/2 \end{cases}$<br>and<br>$x[n+N] = x[n]$ | $2\pi\sum_{k=-\infty}^{+\infty} a_k\delta\left(\omega - \frac{2\pi k}{N}\right)$ | $a_k = \frac{\sin[(2\pi k/N)(N_1 + \frac{1}{2})]}{N\sin[2\pi k/2N]}$, $k \neq 0, \pm N, \pm 2N, \ldots$<br>$a_k = \frac{2N_1 + 1}{N}$, $k = 0, \pm N, \pm 2N, \ldots$ |
| $\sum_{k=-\infty}^{+\infty}\delta[n - kN]$ | $\frac{2\pi}{N}\sum_{k=-\infty}^{+\infty}\delta\left(\omega - \frac{2\pi k}{N}\right)$ | $a_k = \frac{1}{N}$ for all $k$ |
| $a^n u[n]$, $\lvert a\rvert < 1$ | $\frac{1}{1 - ae^{-j\omega}}$ | — |
| $x[n]\begin{cases} 1, & \lvert n\rvert \leq N_1 \\ 0, & \lvert n\rvert > N_1 \end{cases}$ | $\frac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$ | — |
| $\frac{\sin Wn}{\pi n} = \frac{W}{\pi}\operatorname{sinc}\left(\frac{Wn}{\pi}\right)$<br>$0 < W < \pi$ | $X(\omega) = \begin{cases} 1, & 0 \leq \lvert\omega\rvert \leq W \\ 0, & W < \lvert\omega\rvert \leq \pi \end{cases}$<br>$X(\omega)$ periodic with period $2\pi$ | — |
| $\delta[n]$ | 1 | — |
| $u[n]$ | $\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{+\infty}\pi\delta(\omega - 2\pi k)$ | — |
| $\delta[n - n_0]$ | $e^{-j\omega n_0}$ | — |
| $(n+1)a^n u[n]$, $\lvert a\rvert < 1$ | $\frac{1}{(1 - ae^{-j\omega})^2}$ | — |
| $\frac{(n+r-1)!}{n!(r-1)!}a^n u[n]$, $\lvert a\rvert < 1$ | $\frac{1}{(1 - ae^{-j\omega})^r}$ | — |

# Summary

- **Discrete-Time Fourier Transform**
  - Fourier transform and inverse Fourier transform for aperiodic signals
  - Fourier transform for periodic signals, relationship with Fourier coefficients

- **Properties of DT Fourier Transform**